

on account of (F.4), (F.5), thus also

$$d \log p_t(X_t) = -\operatorname{div}(v_t(X_t)) dt, \quad 0 \leq t \leq 1. \quad (\text{F.7})$$

Recall now the function $q(x) = e^{-2\Psi(x)}$, for which

$$d \log q(X_t) = -\langle 2 \nabla \Psi(X_t), dX_t \rangle_{\mathbb{R}^n} = -\langle 2 \nabla \Psi(X_t), v_t(X_t) \rangle_{\mathbb{R}^n} dt. \quad (\text{F.8})$$

For the likelihood ratio function $\ell_t(\cdot)$ of (3.64) we get from (F.7) and (F.8) that

$$d \log \ell_t(X_t) = \langle 2 \nabla \Psi(X_t), v_t(X_t) \rangle_{\mathbb{R}^n} dt - \operatorname{div}(v_t(X_t)) dt, \quad 0 \leq t \leq 1. \quad (\text{F.9})$$

Taking expectations in the integral version of (F.9) we obtain that the difference

$$H(P_t | Q) - H(P_0 | Q) = \mathbb{E}_\nu[\log \ell_t(X_t)] - \mathbb{E}_\nu[\log \ell_0(X_0)] \quad (\text{F.10})$$

is equal to

$$\mathbb{E}_\nu \left[\int_0^t \left(\langle 2 \nabla \Psi(X_\theta), v_\theta(X_\theta) \rangle_{\mathbb{R}^n} - \operatorname{div}(v_\theta(X_\theta)) \right) d\theta \right] \quad (\text{F.11})$$

for $t \in [0, 1]$. Consequently,

$$\lim_{t \downarrow 0} \frac{H(P_t | Q) - H(P_0 | Q)}{t} = \mathbb{E}_\nu \left[\langle 2 \nabla \Psi(X_0), v_0(X_0) \rangle_{\mathbb{R}^n} - \operatorname{div}(v_0(X_0)) \right]. \quad (\text{F.12})$$

Integrating by parts, we see that

$$\mathbb{E}_\nu[\operatorname{div}(v_0(X_0))] = \int_{\mathbb{R}^n} \operatorname{div}(v_0(x)) p_0(x) dx = - \int_{\mathbb{R}^n} \langle v_0(x), \nabla p_0(x) \rangle_{\mathbb{R}^n} dx \quad (\text{F.13})$$

$$= -\langle \nabla \log p_0(X_0), v_0(X_0) \rangle_{L^2(\nu)}. \quad (\text{F.14})$$

Recalling (F.12) and combining it with the relation $\nabla \log \ell_t(x) = \nabla \log p_t(x) + 2 \nabla \Psi(x)$, as well as with (F.13) and (F.14), we get

$$\lim_{t \downarrow 0} \frac{H(P_t | Q) - H(P_0 | Q)}{t} = \langle \nabla \log \ell_0(X_0), v_0(X_0) \rangle_{L^2(\nu)}. \quad (\text{F.15})$$

Since $v_0 = \gamma$, this leads to (3.65). $\square$

G. TIME REVERSAL OF DIFFUSIONS

We review in the present section the theory of time reversal for diffusion processes developed by Föllmer [Föl85, Föl86], Haussmann and Pardoux [HP86], and Pardoux [Par86]. This section can be read independently of the rest of the paper; it does not present novel results.

G.1. INTRODUCTION

It is very well known that the Markov property is invariant under time reversal. In other words, a Markov process remains a Markov process under time reversal (e.g., [RW00a] Exercise E60.41, p. 162). On the other hand, it is also well known that the strong Markov property is not necessarily preserved under time reversal (e.g., [RW00a] p. 330), and neither is the semimartingale property (e.g., [Wal82]). The reason for such failure is the same in both cases: after reversing time, “we may know too much”. Thus, the following questions arise rather naturally:

Given a diffusion process (in particular, a strong Markov semimartingale with continuous paths) $X = (X(t))_{0 \leq t \leq T}$ with certain specific drift and dispersion characteristics, under what conditions might the time-reversed process

$$\hat{X}(s) := X(T - s), \quad 0 \leq s \leq T, \quad (\text{G.1})$$

also be a diffusion? if it happens to be, what are the characteristics of the time-reversed diffusion?

Such questions go back at least to Boltzmann [Bol96, Bol98a, Bol98b], Schrödinger [Sch31, Sch32] and Kolmogorov [Kol37]; they were dealt with systematically by Nelson [Nel01] (see also Carlen [Car84]) in the context of Nelson's dynamical theories for Brownian motion and diffusion. There is now a rather complete theory that answers these questions and provides, as a kind of “bonus”, some rather unexpected results as well. It was developed in the context of theories of filtering, interpolation and extrapolation, where such issues arise naturally — most notably Haussmann and Pardoux [HP86], and Pardoux [Par86]. Very interesting related results in a non-Markovian context, but with dispersion structure given by the identity matrix, have been obtained by Föllmer [Föl85, Föl86]. Let us refer also to the papers [Nag64, NM79] dealing with time reversal of Markov processes, and to the book [Nag93] on diffusion theory. In what follows, this theory is presented in the spirit of the expository paper by Meyer [Mey94].

G.2. THE SETTING

We place ourselves on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$, $\mathbb{F} = (\mathcal{F}(t))_{0 \leq t \leq T}$ rich enough to support an $\mathbb{R}^d$ -valued Brownian motion $W = (W_1, \dots, W_d)'$ adapted to $\mathbb{F}$, as well as an independent $\mathcal{F}(0)$ -measurable random vector $\xi = (\xi_1, \dots, \xi_n)': \Omega \rightarrow \mathbb{R}^n$. In fact, we shall assume that $\mathbb{F}$ is the filtration generated by these two objects, in the sense that we shall take

$$\mathcal{F}(t) = \sigma(\xi, W(\theta): 0 \leq \theta \leq t), \quad 0 \leq t \leq T,$$

modulo $\mathbb{P}$ -augmentation. Next, we assume that the system of stochastic equations

$$X_i(t) = \xi_i + \int_0^t b_i(\theta, X(\theta)) d\theta + \sum_{\nu=1}^d \int_0^t s_{i\nu}(\theta, X(\theta)) dW_\nu(\theta), \quad 0 \leq t \leq T, \quad (\text{G.2})$$

for $i = 1, \dots, n$ admits a pathwise unique, strong solution. It is then well known that the resulting continuous process $X = (X_1, \dots, X_n)'$ is $\mathbb{F}$ -adapted (the strong solvability of the equation (G.2)), which implies that we have also

$$\mathcal{F}(t) = \sigma(X(\theta), W(\theta): 0 \leq \theta \leq t) = \sigma(X(0), W(t) - W(0): 0 \leq \theta \leq t) \quad (\text{G.3})$$

modulo $\mathbb{P}$ -augmentation, for $0 \leq t \leq T$; as well as that X has the strong Markov property, and is thus a diffusion process with drifts $b_i(\cdot, \cdot)$ and dispersions $s_{i\nu}(\cdot, \cdot)$, $i = 1, \dots, n$, $\nu = 1, \dots, d$. We shall denote the $(i, j)^{\text{th}}$ entry of the covariance matrix $a(t, x) := s(t, x) s'(t, x)$ by

$$a_{ij}(t, x) := \sum_{\nu=1}^d s_{i\nu}(t, x) s_{j\nu}(t, x), \quad 1 \leq i, j \leq n.$$

These characteristics are given mappings from $[0, T] \times \mathbb{R}^n$ into $\mathbb{R}$ with sufficient smoothness; in particular, such that the probability density function $p(t, \cdot): \mathbb{R}^n \rightarrow (0, \infty)$ in

$$\mathbb{P}[X(t) \in A] = \int_A p(t, x) dx, \quad A \in \mathcal{B}(\mathbb{R}^n),$$

is smooth. Sufficient conditions on the drift $b_i(\cdot, \cdot)$ and dispersion $s_{i\nu}(\cdot, \cdot)$ characteristics that lead to such smoothness, are provided by the Hörmander hypoellipticity conditions; see for instance

[Bel95], [Nua06] for this result, as well as [Rog85] for a very simple argument in the one-dimensional case ($n = d = 1$), and to the case of Langevin type equation (2.1) for arbitrary $n \in \mathbb{N}$. We refer to [Fri75], [RW00b] or [KS98] for the basics of the theory of stochastic equations of the form (G.2)

The probability density function $p(t, \cdot) : \mathbb{R}^n \rightarrow (0, \infty)$ solves the forward Kolmogorov [Kol31] equation [Fri75, p. 149]

$$\partial_t p(t, x) = \frac{1}{2} \sum_{i,j=1}^n D_{ij}^2(a_{ij}(t, x) p(t, x)) - \sum_{i=1}^n D_i(b_i(t, x) p(t, x)), \quad (t, x) \in (0, T] \times \mathbb{R}^n. \quad (\text{G.4})$$

If the drift and dispersion characteristics do not depend on time, and an invariant probability measure exists for the diffusion process of (G.2) the density function $p(\cdot)$ of this measure solves the stationary version of this forward Kolmogorov equation, to wit

$$\frac{1}{2} \sum_{i,j=1}^n D_{ij}^2(a_{ij}(x) p(x)) = \sum_{i=1}^n D_i(b_i(x) p(x)), \quad x \in \mathbb{R}^n. \quad (\text{G.5})$$

G.3. TIME REVERSAL AND THE BACKWARDS FILTRATION

Consider now the family of σ -algebras $(\hat{\mathcal{F}}(t))_{0 \leq t \leq T}$ given by

$$\hat{\mathcal{F}}(t) := \sigma(X(\theta), W(\theta) - W(t) : t \leq \theta \leq T), \quad 0 \leq t \leq T. \quad (\text{G.6})$$

It is not hard to see that the σ -algebra in (G.6) is expressed equivalently as

$$\begin{aligned} \hat{\mathcal{F}}(t) &= \sigma(X(t), W(\theta) - W(t) : t \leq \theta \leq T) = \sigma(X(t), W(\theta) - W(T) : t \leq \theta \leq T) \\ &= \sigma(X(T), W(\theta) - W(t) : t \leq \theta \leq T) = \sigma(X(T)) \vee \mathcal{H}(t). \end{aligned} \quad (\text{G.7})$$

Here, the σ -algebra generated by the Brownian increments after time t , namely,

$$\mathcal{H}(t) := \sigma(W(\theta) - W(t) : t \leq \theta \leq T), \quad 0 \leq t \leq T, \quad (\text{G.8})$$

is independent of the random vector $X(t)$. The time-reversed processes $\hat{X}$ as in (G.1) as well as

$$\widetilde{W}(s) := W(T - s) - W(T), \quad 0 \leq s \leq T, \quad (\text{G.9})$$

are both adapted to the *backwards filtration* $\hat{\mathbb{F}} := (\hat{\mathcal{F}}(T - s))_{0 \leq s \leq T}$, where

$$\begin{aligned} \hat{\mathcal{F}}(T - s) &= \sigma(X(T - u), W(T - u) - W(T - s) : 0 \leq u \leq s) \\ &= \sigma(\hat{X}(u), \widetilde{W}(u) - \widetilde{W}(s) : 0 \leq u \leq s) \end{aligned} \quad (\text{G.10})$$

from (G.6). Note that, by complete analogy with (G.3), we have also

$$\hat{\mathcal{F}}(T - s) = \sigma(X(T), W(T - u) - W(T - s) : 0 \leq u \leq s) = \sigma(\hat{X}(0)) \vee \mathcal{H}(T - s) \quad (\text{G.11})$$

on account of (G.7) where

$$\mathcal{H}(T - s) = \sigma(W(T - u) - W(T - s) : 0 \leq u \leq s) = \sigma(\widetilde{W}(u) - \widetilde{W}(s) : 0 \leq u \leq s). \quad (\text{G.12})$$

In words: the σ -algebra $\hat{\mathcal{F}}(T - s)$ is generated by the terminal value $X(T)$ of the forward process (i.e., by the original value $\hat{X}(0)$ of the backward process) and by the increments of the time-reversed process $\widetilde{W}$ on $[0, s]$; see the expressions right above. Furthermore, the σ -algebra $\hat{\mathcal{F}}(T - s)$ measures all the random variables $\hat{X}(u)$, $u \in [0, s]$.

Remark G.1. In fact, the time-reversed process $\widetilde{W}$ is a Brownian motion of the backwards filtration $\mathbb{H} := (\mathcal{H}(T - s))_{0 \leq s \leq T} \subseteq \hat{\mathbb{F}}$ as in (G.12) generated by the increments of W after time $T - s$, $0 \leq s \leq T$. This is because it is a martingale of this filtration, has continuous paths, and its quadratic variation is that of Brownian motion (Lévy's theorem [KS98 Theorem 5.1]). In the next subsection we shall see that the process $\widetilde{W}$ is only a semimartingale of the larger backwards filtration $\hat{\mathbb{F}} = (\hat{\mathcal{F}}(T - s))_{0 \leq s \leq T}$, and identify its semimartingale decomposition.

G.4. SOME REMARKABLE BROWNIAN MOTIONS

Following the exposition and ideas in [Mey94](#), we start with a couple of observations. First, for every $t \in [0, T]$ and every integrable, $\widehat{\mathcal{F}}(t)$ -measurable random variable $\mathcal{K}$, we have

$$\mathbb{E}[\mathcal{K} | \mathcal{F}(t)] = \mathbb{E}[\mathcal{K} | X(t)], \quad \text{almost surely.} \quad (\text{G.13})$$

Secondly, we fix a function $G \in \mathcal{C}_0^\infty(\mathbb{R}^n)$ and a time-point $t \in (0, T]$, and define

$$g(\theta, x) := \mathbb{E}[G(X(t)) | X(\theta) = x], \quad (\theta, x) \in [0, t] \times \mathbb{R}^n.$$

Invoking the Markov property of X , we deduce that the process

$$g(\theta, X(\theta)) = \mathbb{E}[G(X(t)) | X(\theta)] = \mathbb{E}[G(X(t)) | \mathcal{F}(\theta)], \quad 0 \leq \theta \leq t$$

is an $\mathbb{F}$ -martingale, and obtain

$$G(X(t)) - g(\theta, X(\theta)) = g(t, X(t)) - g(\theta, X(\theta)) = \sum_{i=1}^n \sum_{\nu=1}^d \int_{\theta}^t D_i g(v, X(v)) s_{i\nu}(v, X(v)) dW_{\nu}(v).$$

For every index $\nu = 1, \dots, d$ this gives, after integrating by parts,

$$\begin{aligned} \mathbb{E}[(W_{\nu}(t) - W_{\nu}(\theta)) \cdot G(X(t))] &= \mathbb{E}[(W_{\nu}(t) - W_{\nu}(\theta)) \cdot (g(t, X(t)) - g(\theta, X(\theta)))] \\ &= \mathbb{E}\left[\sum_{i=1}^n \int_{\theta}^t D_i g(v, X(v)) s_{i\nu}(v, X(v)) dv\right] = \sum_{i=1}^n \int_{\theta}^t \int_{\mathbb{R}^n} (D_i g \cdot s_{i\nu})(v, x) p(v, x) dx dv \\ &= - \sum_{i=1}^n \int_{\theta}^t \int_{\mathbb{R}^n} g(v, x) D_i (p(v, x) s_{i\nu}(v, x)) dx dv = - \int_{\theta}^t \int_{\mathbb{R}^n} g(v, x) \operatorname{div} (p(v, x) \bar{s}_{\nu}(v, x)) dx dv \\ &= - \int_{\theta}^t \mathbb{E}\left[g(v, X(v)) \cdot \frac{\operatorname{div}(p \bar{s}_{\nu})}{p}(v, X(v))\right] dv = - \mathbb{E}\left[G(X(t)) \cdot \int_{\theta}^t \frac{\operatorname{div}(p \bar{s}_{\nu})}{p}(v, X(v)) dv\right]. \end{aligned}$$

Here $\bar{s}_{\nu}(v, \cdot)$ is the ν^{th} column vector of the dispersion matrix. Comparing the first and last expressions in the above string of equalities, we see that with $0 \leq \theta \leq t$ we have

$$\mathbb{E}\left[G(X(t)) \cdot \left(W_{\nu}(t) - W_{\nu}(\theta) + \int_{\theta}^t \frac{\operatorname{div}(p \bar{s}_{\nu})}{p}(v, X(v)) dv\right)\right] = 0 \quad (\text{G.14})$$

for every $G \in \mathcal{C}_0^\infty(\mathbb{R}^n)$, and thus by extension for every bounded, measurable $G: \mathbb{R}^n \rightarrow \mathbb{R}$.

Theorem G.2. *The vector process $B = (B_1, \dots, B_d)'$ defined as*

$$B_{\nu}(s) := \widetilde{W}_{\nu}(s) - \int_0^s \frac{\operatorname{div}(p \bar{s}_{\nu})}{p}(T - u, \widehat{X}(u)) du \quad (\text{G.15})$$

$$= W_{\nu}(T - s) - W_{\nu}(T) - \int_{T-s}^T \frac{\operatorname{div}(p \bar{s}_{\nu})}{p}(v, X(v)) dv, \quad 0 \leq s \leq T, \quad (\text{G.16})$$

for $\nu = 1, \dots, d$, is a Brownian motion of the backwards filtration $\widehat{\mathbb{F}} = (\widehat{\mathcal{F}}(T - s))_{0 \leq s \leq T}$.

Remark G.3. The Brownian motion process B is thus independent of $\widehat{\mathcal{F}}(T)$, and therefore also of the $\widehat{\mathcal{F}}(T)$ -measurable random variable $X(T)$. A bit more generally,

$$\{B(T - \theta) - B(T - t): 0 \leq \theta \leq t\} \quad \text{is independent of} \quad \widehat{\mathcal{F}}(t) \supseteq \sigma(X(v): t \leq v \leq T).$$

Note also from [\(G.16\)](#) that

$$B_{\nu}(T - \theta) - B_{\nu}(T - t) = W_{\nu}(\theta) - W_{\nu}(t) - \int_{\theta}^t \frac{\operatorname{div}(p \bar{s}_{\nu})}{p}(v, X(v)) dv, \quad 0 \leq \theta \leq t.$$

Reversing time once again, we obtain the following corollary of [Theorem G.2](#)

Corollary G.4. *The $\mathbb{F}$ -adapted vector process $V = (V_1, \dots, V_d)'$ with components*

$$V_\nu(t) := B_\nu(T - t) - B_\nu(T) = W_\nu(t) + \int_0^t \frac{\operatorname{div}(p \bar{s}_\nu)}{p}(v, X(v)) \, dv, \quad 0 \leq t \leq T, \quad (\text{G.17})$$

for $\nu = 1, \dots, d$, is yet another Brownian motion (with respect to its own filtration $\mathbb{F}^V \subseteq \mathbb{F}$). This process is independent of the random variable $X(T)$; and a bit more generally, for every $t \in (0, T]$, the σ -algebra

$$\mathcal{F}^V(t) := \sigma(V(\theta) : 0 \leq \theta \leq t) \quad (\text{G.18})$$

generated by present-and-past values of V , is independent of $\sigma(X(v) : t \leq v \leq T)$, the σ -algebra generated by present-and-future values of X .

Proof of [Theorem G.2](#). It suffices to show that each component process B_ν is a martingale of the backwards filtration $\hat{\mathbb{F}}$; because then, in view of the continuity of paths and the easily checked property $\langle B_\nu, B_\ell \rangle(s) = s \delta_{\nu\ell}$, we can deduce that each B_ν is a Brownian motion in the backwards filtration $\hat{\mathbb{F}}$ (and of course also in its own filtration), and that B_ν, B_ℓ are independent for $\ell \neq \nu$, by appealing to Lévy's theorem once again.

Now we have to show $\mathbb{E}[(B_\nu(T - \theta) - B_\nu(T - t)) \cdot \mathcal{K}] = 0$ for $0 \leq \theta \leq t \leq T$ and every bounded, $\hat{\mathcal{F}}(t)$ -measurable $\mathcal{K}$; equivalently,

$$\mathbb{E} \left[\mathbb{E}[\mathcal{K} | \mathcal{F}(t)] \cdot \left(W_\nu(t) - W_\nu(\theta) + \int_\theta^t \frac{\operatorname{div}(p \bar{s}_\nu)}{p}(v, X(v)) \, dv \right) \right] = 0,$$

as the expression inside the curved braces is $\mathcal{F}(t)$ -measurable. But recalling [\(G.13\)](#) we have that $\mathbb{E}[\mathcal{K} | \mathcal{F}(t)] = \mathbb{E}[\mathcal{K} | X(t)] = G(X(t))$ for some bounded, measurable $G: \mathbb{R}^n \rightarrow \mathbb{R}$, and the desired result follows from [\(G.14\)](#). $\square$

G.5. THE DIFFUSION PROPERTY UNDER TIME REVERSAL

Let us return now to the question, whether the time-reversed process $\hat{X}$ of [\(G.1\)](#), [\(G.2\)](#) is a diffusion. We start by expressing X_i of [\(G.2\)](#) in terms of a backwards Itô integral (see [Subsection G.6](#)) as

$$\begin{aligned} X_i(t) - \xi_i - \int_0^t b_i(\theta, X(\theta)) \, d\theta &= \sum_{\nu=1}^d \int_0^t s_{i\nu}(\theta, X(\theta)) \, dW_\nu(\theta) \\ &= \sum_{\nu=1}^d \left(\int_0^t s_{i\nu}(\theta, X(\theta)) \bullet dW_\nu(\theta) - \langle s_{i\nu}(\cdot, X), W_\nu \rangle(t) \right). \end{aligned}$$

From [\(G.2\)](#) we have by Itô's formula that the process

$$s_{i\nu}(\cdot, X) - s_{i\nu}(0, \xi) - \sum_{j=1}^n \sum_{\kappa=1}^d \int_0^\cdot D_j s_{i\nu}(\theta, X(\theta)) \cdot s_{j\kappa}(\theta, X(\theta)) \, dW_\kappa(\theta)$$

is of finite first variation, therefore $\langle s_{i\nu}(\cdot, X), W_\nu \rangle(t) = \sum_{j=1}^n \int_0^t s_{j\nu}(\theta, X(\theta)) \, D_j s_{i\nu}(\theta, X(\theta)) \, d\theta$. We conclude

$$X_i(t) = \xi_i - \int_0^t \left(\sum_{j=1}^n \sum_{\nu=1}^d s_{j\nu} \, D_j s_{i\nu} - b_i \right)(\theta, X(\theta)) \, d\theta + \sum_{\nu=1}^d \int_0^t s_{i\nu}(\theta, X(\theta)) \bullet dW_\nu(\theta).$$

Evaluating also at $t = T$, then subtracting, we obtain

$$X_i(t) = X_i(T) + \int_t^T \left(\sum_{j=1}^n \sum_{\nu=1}^d s_{j\nu} D_j s_{i\nu} - b_i \right) (\theta, X(\theta)) d\theta - \sum_{\nu=1}^d \int_t^T s_{i\nu}(\theta, X(\theta)) \bullet dW_\nu(\theta),$$

as well as

$$\hat{X}_i(s) = \hat{X}_i(0) + \int_0^s \left(\sum_{j=1}^n \sum_{\nu=1}^d s_{j\nu} D_j s_{i\nu} - b_i \right) (T-u, \hat{X}(u)) du + \sum_{\nu=1}^d \int_0^s s_{i\nu}(T-u, \hat{X}(u)) d\widetilde{W}_\nu(u)$$

by reversing time. It is important here to note that the backward Itô integral for W becomes a forward Itô integral for the process $\widetilde{W}$, the time reversal of W in the manner of [\(G.9\)](#).

But now let us recall [\(G.15\)](#) on the strength of which the above expression takes the form

$$\begin{aligned} \hat{X}_i(s) &= \hat{X}_i(0) + \sum_{\nu=1}^d \int_0^s s_{i\nu}(T-u, \hat{X}(u)) dB_\nu(u) \\ &\quad + \int_0^s \left(\sum_{j=1}^n \sum_{\nu=1}^d s_{j\nu} D_j s_{i\nu} + \sum_{\nu=1}^d s_{i\nu} \frac{\operatorname{div}(p \overline{s}_\nu)}{p} - b_i \right) (T-u, \hat{X}(u)) du, \quad 0 \leq s \leq T. \end{aligned}$$

But in conjunction with [Theorem G.2](#) this means that the time-reversed process $\hat{X}$ of [\(G.1\)](#), [\(G.2\)](#) is a semimartingale of the backwards filtration $\hat{\mathbb{F}} = (\hat{\mathcal{F}}(T-s))_{0 \leq s \leq T}$, with decomposition

$$\hat{X}_i(s) = \hat{X}_i(0) + \int_0^s \hat{b}_i(T-u, \hat{X}(u)) du + \sum_{\nu=1}^d \int_0^s s_{i\nu}(T-u, \hat{X}(u)) dB_\nu(u) \quad (\text{G.19})$$

for $0 \leq s \leq T$, where, for each $i = 1, \dots, n$, the function $\hat{b}_i(\cdot, \cdot)$ is specified by

$$\begin{aligned} \hat{b}_i(t, x) + b_i(t, x) &= \sum_{j=1}^n \sum_{\nu=1}^d s_{j\nu}(t, x) D_j s_{i\nu}(t, x) + \sum_{\nu=1}^d s_{i\nu}(t, x) \frac{\operatorname{div}(p(t, x) \overline{s}_\nu(t, x))}{p(t, x)} \\ &= \sum_{j=1}^n \sum_{\nu=1}^d s_{j\nu}(t, x) D_j s_{i\nu}(t, x) + \sum_{\nu=1}^d \frac{s_{i\nu}(t, x)}{p(t, x)} \left(\sum_{j=1}^n D_j (p(t, x) s_{j\nu}(t, x)) \right) \\ &= \sum_{j=1}^n (D_j a_{ij}(t, x) + a_{ij}(t, x) \cdot D_j \log p(t, x)). \end{aligned}$$

Theorem G.5. *Under the assumptions of this section, the time-reversed process $\hat{X}$ of [\(G.1\)](#), [\(G.2\)](#) is a diffusion in the backwards filtration $\hat{\mathbb{F}} = (\hat{\mathcal{F}}(T-s))_{0 \leq s \leq T}$, with characteristics as in [\(G.19\)](#), namely, dispersions $s_{i\nu}(T-s, x)$ and drifts $\hat{b}_i(T-s, x)$ given by the generalized Nelson equation*

$$\hat{b}_i(t, x) + b_i(t, x) = \sum_{j=1}^n \left(D_j a_{ij}(t, x) + a_{ij}(t, x) \cdot D_j \log p(t, x) \right), \quad i = 1, \dots, n. \quad (\text{G.20})$$

Equivalently, and with $\operatorname{div}(a(t, x)) := (\sum_{j=1}^n D_j a_{ij}(t, x))_{1 \leq i \leq n}$, we write

$$\hat{b}(t, x) + b(t, x) = \operatorname{div}(a(t, x)) + a(t, x) \cdot \nabla \log p(t, x). \quad (\text{G.21})$$

Remark G.6. This result can be extended to the case where the sums of the distributional derivatives $\sum_{j=1}^n D_j (a_{ij}(t, x) p(t, x))$, $i = 1, \dots, n$, are only assumed to be locally integrable functions of $x \in \mathbb{R}^n$; see [\[MNS89, RVW01\]](#).

Remark G.7 (Some filtration comparisons). For an invertible dispersion matrix $s(\cdot, \cdot)$, it follows from (G.19) that the Brownian motion B is adapted to the filtration generated by $\hat{X}$; that is,

$$\mathcal{F}^B(s) \subseteq \mathcal{F}^{\hat{X}}(s) := \sigma(\hat{X}(u): 0 \leq u \leq s) = \sigma(X(T-u): 0 \leq u \leq s), \quad 0 \leq s \leq T. \quad (\text{G.22})$$

Now recall (G.15) in its light, the filtration comparison in (G.22) implies $\mathcal{F}^{\tilde{W}}(s) \subseteq \mathcal{F}^{\hat{X}}(s)$, thus $\mathcal{H}(T-s) \subseteq \mathcal{F}^{\tilde{W}}(s) \subseteq \mathcal{F}^{\hat{X}}(s)$ from (G.12) for $0 \leq s \leq T$, and from (G.11) also

$$\hat{\mathcal{F}}(T-s) \subseteq \mathcal{F}^{\hat{X}}(s), \quad 0 \leq s \leq T. \quad (\text{G.23})$$

But we have also the reverse inclusion $\mathcal{F}^{\hat{X}}(s) \subseteq \hat{\mathcal{F}}(T-s)$ on account of (G.10) and (G.22); therefore, $\mathcal{F}^{\hat{X}}(s) = \hat{\mathcal{F}}(T-s)$ holds for all $0 \leq s \leq T$ when $s(\cdot, \cdot)$ is invertible. These considerations inform our choice of backwards filtration $\mathcal{G}(T-s) \equiv \mathcal{F}^{\hat{X}}(s)$, $0 \leq s \leq T$, in (3.22)

G.6. THE BACKWARDS ITÔ INTEGRAL

For two continuous semimartingales $X = X(0) + M + B$ and $Y = Y(0) + N + C$, with B, C continuous adapted processes of finite variation and M, N continuous local martingales, let us recall the definition of the Fisk-Stratonovich integral in [KS98, Definition 3.3.13, p. 156], as well as its properties in [KS98, Problem 3.3.14] and [KS98, Problem 3.3.15].

By analogy with this definition, we introduce the *backwards Itô integral*

$$\int_0^\cdot Y(t) \bullet dX(t) := \int_0^\cdot Y(t) dM(t) + \int_0^\cdot Y(t) dB(t) + \langle M, N \rangle, \quad (\text{G.24})$$

where the first (respectively, the second) integral on the right-hand side is to be interpreted in the Itô (respectively, the Lebesgue-Stieltjes) sense.

If $\Pi = \{t_0, t_1, \dots, t_m\}$ is a partition of the interval $[0, T]$ with $0 = t_0 < t_1 < \dots < t_m = T$, then the sums

$$\sum_{j=0}^{m-1} Y(t_{j+1})(X(t_{j+1}) - X(t_j)) \quad (\text{G.25})$$

converge in probability to $\int_0^T Y(t) \bullet dX(t)$ as the mesh $\|\Pi\|$ of the partition tends to zero. Note that the increments of X here “stick backwards into the past”, as opposed to “sticking forward into the future” as in the Itô integral.

For the backwards Itô integral we have the change of variable formula

$$f(X) = f(X(0)) + \sum_{i=1}^n \int_0^\cdot D_i f(X(t)) \bullet dX_i(t) - \frac{1}{2} \sum_{i,j=1}^n \int_0^\cdot D_{ij}^2 f(X(t)) d\langle M_i, M_j \rangle(t), \quad (\text{G.26})$$

where now $X = (X_1, \dots, X_n)'$ is a vector of continuous semimartingales $X_1, \dots, X_n$ of the form $X_i = X_i(0) + M_i + B_i$ as above, for $i = 1, \dots, n$. Note the change of sign, from $(+)$ to $(-)$ in the last, stochastic correction term.